

The Policy (π) 🧠

In Reinforcement Learning, we need a formal way to define *how* an agent chooses an action based on its current situation. This function is called the **Policy**.

Definition

A Policy, denoted by the Greek letter **Pi** (π), is a function that maps a **State** (s) to an **Action** (a).

$$a = \pi(s)$$

- **Input:** The current State s .
- **Output:** The Action a that the agent should take.

The Goal of Reinforcement Learning

The ultimate goal of any RL algorithm is to find the **Optimal Policy** (often denoted as π^*).

- We are not just looking for *any* policy.
- We want to find the specific policy that, when followed, results in the **maximum possible Return** (sum of discounted rewards).

Example: Mars Rover Policy

Imagine a policy designed to grab the nearest reward:

Current State (s)	Policy Output $\pi(s)$	Meaning
State 2	Left	"If I am in State 2, go Left."
State 3	Left	"If I am in State 3, go Left."
State 4	Left	"If I am in State 4, go Left."
State 5	Right	"If I am in State 5, go Right."

In this example, the policy π is simply a lookup table or a set of rules telling the robot exactly what to do in every possible location.

Terminology Note

- In control theory or robotics, this might be called a "**Controller**."
- In Reinforcement Learning, the standard term is always "**Policy**."